

Post-Compromise Security for Physical AI: Deterministic Runtime Enforcement of Physical Action Authority in Autonomous UAVs

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Abstract—Autonomous Physical AI systems increasingly connect learned or language-based controllers to cyber-physical actuators. This creates a security problem that differs from conventional model robustness: when a Physical AI controller is fully compromised, where does the last enforceable security boundary remain? We formulate the Post-Compromise Physical Action Authority Problem and investigate whether execution authority can be placed at the transition from autonomous intent to physical action, bounding action authority independently of cognitive controller integrity.

We present DROS-Kinetic, a reference implementation that places a deterministic authorization gate between an autonomous control process and the flight-control command interface. The design separates cognitive compromise from physical execution authority through explicit action, principal, capability, and policy checks. We formalize three core properties: Unauthorized Command-Induced Actuation Invariance (UCIAI) (an enforcement property), Conditional Empirical Safety-Envelope Preservation (a conditional cyber-physical system property), and Delegation Non-Escalation (a multi-agent governance property) under explicitly stated vehicle, dynamic, and disturbance assumptions.

We evaluate the system using a six-stage experimental ladder (T1–T6) covering nominal execution, adversarial controller compromise, execution containment, physical safety-envelope enforcement, multi-agent delegation, and revocation under boundary-oriented stress workloads. Results are reported in terms of unauthorized-actuation containment, safety-envelope violations, delegation propagation depth, and decomposed latency layers ($L_{\text{enforcement}} \ll L_{\text{transport}} \ll L_{\text{physical}}$). The study provides an experimentally testable basis for evaluating post-compromise security at the boundary between autonomous cognition and physical actuation.

Index Terms—Physical AI, Autonomous UAVs, Post-Compromise Security, Runtime Enforcement, Execution Authority, Cyber-Physical Systems.

I. Introduction

The rapid convergence of Large Language Models (LLMs), Vision-Language-Action (VLA) models, and embodied robotics has accelerated the deployment of autonomous Physical AI agents in high-stakes domains, ranging from unmanned aerial vehicles (UAVs) and industrial manipulators to autonomous transport systems. However, placing unconstrained generative or learned reasoning systems in direct control of physical actuators introduces an existential safety assurance paradox. When the cognitive controller encounters adversarial sensory inputs, prompt

injection attacks, or zero-day logic subversion, text-level guardrails within the model substrate fail to prevent the generation of catastrophic actuator commands.

In traditional software systems, a security breach results in unauthorized data exfiltration or state mutation, which can often be mitigated via transactional rollbacks or cryptographic revocations. In contrast, Physical AI systems possess physical mass, kinetic momentum, and mechanical dynamics. A single unauthorized command—such as an in-flight motor disarm or a high-velocity dive into restricted airspace—leads to irreversible physical destruction and human safety hazards.

This fundamental reality leads to the central question of this work:

When a Physical AI controller is fully compromised,
where does the last enforceable security boundary
remain?

We investigate whether that boundary can be placed at the transition from autonomous intent to physical action, bounding action authority independently of cognitive controller integrity. We formulate the central security thesis:

$$\boxed{\begin{array}{l} \text{Compromise(Cognitive Controller)} \\ \not\Rightarrow \text{Compromise(Physical Execution Authority)} \end{array}} \quad (1)$$

To rigorously study this problem, this paper introduces DROS-Kinetic, an open-source reference implementation positioned at the boundary between the autonomous controller and the flight-control hardware interface. Using autonomous UAVs as a controlled physical platform, we make the following contributions:

- 1) **Problem Formulation:** We formulate the Post-Compromise Physical Action Authority Problem, explicitly decoupling cognitive intent generation from the authority to induce physical actuation.
- 2) **Formal Security Properties:** We formalize three testable properties: Unauthorized Command-Induced Actuation Invariance (UCIAI), Conditional Empirical Safety-Envelope Preservation, and Delegation Non-Escalation under declared disturbance bounds.

- 3) Reference Architecture: We present the DROS-Kinetic execution substrate, decoupling enforcement latency from transport and physical dynamics ($L_{\text{enforcement}} \ll L_{\text{transport}} \ll L_{\text{physical}}$).
- 4) Reproducible Empirical Methodology: We construct a six-tier experimental ladder (T1–T6) and evaluate the system against boundary-oriented adversarial mutations and swarm delegation propagation.

II. Related Work

A. UAV Cybersecurity and Runtime Assurance

Traditional UAV cybersecurity research has heavily focused on communication link encryption, GPS spoofing mitigation, and intrusion detection systems on MAVLink telemetry buses [1]. In parallel, runtime assurance (RTA) frameworks, such as simplex architectures, switch control to a verified baseline controller upon detecting safety violations [2]. However, traditional RTA assumes that the primary controller fails due to software bugs or domain shifts, rather than operating under active, adversarial post-compromise control with legitimate credential misuse.

B. Agentic AI Governance and Physical Embodiment

Recent efforts in LLM agent governance have focused on tool-use containment, capability attenuation, and execution passports in digital environments [3]. When extending agents to Physical AI and robotics, emerging industry standards (e.g., Anthropic Model Hardware Interface) highlight the critical need for hardware-level boundaries. However, existing proposals largely treat safety as an internal alignment problem. Our work addresses the gap where the cognitive controller is already fully compromised, establishing an independent in-band execution authority gate.

C. Research Gap and Experimental Substrate

Existing UAV and Physical AI research provides increasingly mature physical testbeds for autonomous flight, cyber-physical evaluation, and adversarial experimentation. However, the availability of a physical platform alone does not establish an independently verifiable execution-authority boundary between a compromised autonomous controller and the physical actuation layer.

This distinction is particularly important in the post-compromise setting. Once the cognitive controller is assumed to be fully compromised, evaluating whether an attack can influence vehicle behavior is insufficient to determine whether physical execution authority remains constrained. A post-compromise evaluation therefore requires an experimental substrate in which the controller, execution-authority enforcement layer, flight-control interface, and resulting physical effects can be observed and measured as separate stages.

In this work, DROS-Kinetic is used as a reference implementation of such an execution-authority boundary.

The implementation is not itself the research object; rather, it provides a concrete and reproducible substrate through which the proposed security properties can be instantiated, attacked, measured, and evaluated under a declared threat model. This separation also permits the evaluation methodology to be applied to alternative enforcement mechanisms and physical platforms.

III. Post-Compromise Physical Action Authority Problem

A. System and Threat Model

We model an autonomous Physical AI agent as consisting of two distinct domains separated by a trusted execution boundary:

- Untrusted Cognitive Controller (C^*): Modeled as an adversarial black box $C^* : \mathcal{I} \rightarrow \mathcal{A}^*$, where $\mathcal{A}^* \subseteq \mathcal{A}_{\text{arbitrary}}$. The attacker is assumed to have full control over high-level reasoning, prompt contexts, mission planning, and credential submission.
- Trusted Execution Substrate: Positioned between C^* and the flight-control hardware. It comprises the DROS-Kinetic enforcement engine, cryptographic capability stores, and the low-level hardware interface.

Assurance Boundary Disclosures: We explicitly assume the physical integrity of the flight-controller firmware, uncompromised low-level hardware timing, and environmental wind disturbances within defined bounds $\mathcal{W}$. Physical JTAG tampering and mechanical actuator failures are out of scope.

IV. Formal Security Properties

A. Vehicle Dynamics and Input Decomposition

The physical vehicle dynamics are governed by:

$$\dot{x}(t) = f(x(t), u(t)) + w(t), \quad w(t) \in \mathcal{W} (\|w(t)\| \leq W_{\max}) \quad (2)$$

where $x(t) \in \mathbb{R}^n$ is the vehicle state vector, and $u(t) \in \mathcal{U}$ is the total control input. We decompose the control input as:

$$u(t) = u_{\text{nominal}}(t) + \Delta u_{\text{cmd}}(t) \quad (3)$$

where $u_{\text{nominal}}(t)$ denotes existing authorized control actions and natural flight stabilization, and $\Delta u_{\text{cmd}}(t)$ denotes the command-induced actuation directly attributable to the evaluated command.

B. Property P1: Unauthorized Command-Induced Actuation Invariance (UCIAI)

$$\boxed{\text{Auth}(a, s, t) = 0 \implies \Delta u_{\text{cmd}}(a, s, t) \equiv 0} \quad (4)$$

Epistemic Definition: P1 is an enforcement property. A denied action request must introduce zero additional actuation into the execution path. Crucially, $\Delta u_{\text{cmd}} = 0$ does not imply $x(t + \Delta t) = x(t)$, as natural inertia, gravity, and pre-existing aerodynamic momentum continue to govern vehicle dynamics.

C. Property P2: Conditional Empirical Safety-Envelope Preservation

Let $\mathcal{S}_{\text{safe}} \subset \mathcal{X}$ be the defined safe reachability envelope (P2-S). Within a declared operating envelope $\Omega = \{x, u, w, \epsilon_{\text{est}}, \tau\}$, the system empirically demonstrates (P2-E):

$$x(t) \in \mathcal{S}_{\text{safe}} \quad \forall t \in [t_0, t_0 + T] \quad (5)$$

Epistemic Definition: P2 is a conditional cyber-physical empirical property, verified under explicit vehicle modeling and disturbance bounds.

D. Property P3: Delegation Non-Escalation

For parent node i delegating to child node j :

$$\text{Authority}_j \subseteq \text{Attenuate}(\text{Authority}_i) \quad \text{and} \quad \text{Depth} \leq H \quad (6)$$

Any delegation attempt with $\text{Depth} > H$ yields $\text{Auth} = 0 \implies \Delta u_{\text{cmd}} \equiv 0$.

V. DROS-Kinetic Architecture and Latency Decomposition

A. In-Band Enforcement Pipeline

DROS-Kinetic interposes directly between the autonomous planning process and the flight-control communication stack. It enforces bitmask capability checks, dynamic lookahead kinematics, and epoch-based revocation before serializing commands into low-level MAVLink frames.

B. Four-Layer Latency Decomposition

To prevent conflation between binary decision speeds and physical vehicle responses, latency is formally decomposed into four distinct layers:

- L1 (Authorization Primitive): Pure $O(1)$ capability bitmask lookup ($L_1 < 500$ ns).
- L2 (End-to-End Enforcement Decision): In-process C-ABI argument sanitization and kinematic lookahead ($P_{50} = 4.40$ μs , $P_{99} = 13.30$ μs).
- L3 (Command Transport): MAVLink framing and serial/UDP bus delivery (1.20 ms $\sim$ 3.50 ms, transport-dependent).
- L4 (Physical Dynamics Response): Flight controller loop, ESC motor acceleration, and aerodynamic response (50 ms $\sim$ 200 ms, platform-dependent).

$$L_{\text{enforcement}} (L_1/L_2) \ll L_{\text{transport}} (L_3) \ll L_{\text{physical}} (L_4) \quad (7)$$

VI. Cyber-Physical Stopping Horizon Model

In geofence enforcement scenarios, preventing kinetic penetration requires proactive intervention. We model the required stopping horizon d_{safe} as:

$$d_{\text{safe}} = d_{\text{brake}} + d_{\text{latency}} + d_{\text{estimation}} + d_{\text{margin}} \quad (8)$$

where:

- $d_{\text{brake}} = \frac{v_h^2}{2a_{\text{max}}}$ (Kinetic braking distance at horizontal speed v_h and max deceleration a_{max}).
- $d_{\text{latency}} = v_h \times (L_1 + L_2 + L_3) \approx 15.0 \times 0.0035 = 0.05$ m.
- $d_{\text{estimation}} = \sigma_{\text{GNSS}} + \sigma_{\text{VIO}} \approx 2.50$ m (Sensor uncertainty).
- $d_{\text{margin}} = 20.00$ m (Pre-configured aerodynamic safety buffer under wind disturbance W_{max}).

For a nominal test configuration of $v_h = 15.0$ m/s and $a_{\text{max}} = 3.0$ m/s², the predicted intervention threshold is:

$$d_{\text{safe}} = 37.50 + 0.05 + 2.50 + 20.00 = 60.05 \text{ m} \quad (9)$$

VII. Experimental Evaluation

A. T1–T6 Research Protocol

We implement a six-tier experimental protocol executed on high-fidelity Software-in-the-Loop (SITL) quadrotor simulation linked via ROS2 and MAVLink:

- T1 (Nominal): Valid waypoint navigation commands execute with zero trajectory distortion; enforcement overhead is bounded by L_2 .
- T2 (Adversarial Controller): Compromised controller issues arbitrary unauthorized commands (e.g., in-flight Disarm).
- T3 (Execution Containment): Under B0 (Unprotected), in-flight Disarm causes immediate free-fall motor cutoff ($\Delta S > 0$). Under B1 (DROS-Kinetic), unauthorized Disarm is rejected at the C-ABI boundary ($\Delta u_{\text{cmd}} \equiv 0$), preserving stable hover at 25.0m altitude.
- T4 (Safety Envelope): High-speed (15.0 m/s) ingress toward a 500m no-fly-zone (NFZ). Lookahead braking triggers at 61.30 m (predicted 60.05 m), achieving complete halt at +24.60 m prior to the boundary (0.0 m penetration).
- T5 (Swarm Delegation): A 5-hop adversarial delegation propagation attempt is evaluated across a 100-UAV swarm. With policy $H = 2$, Hops 1–2 succeed within attenuated bounds, while Hops 3–5 are deterministically blocked (Zero unauthorized propagation beyond H).
- T6 (Revocation & Stress): Evaluated against a boundary-oriented corpus (NaN, $\pm\text{Inf}$, truncated MAVLink frames, expired epochs) across 10,000 mutations and a 178,000 QPS stress workload. DROS-Kinetic achieved 0 crashes and a 100% fail-closed containment rate ($R_{\text{FC}} = 1.0$), with $P_{99} = 13.30$ μs .

VIII. Discussion and Limitations

A. Assurance Boundary Disclosures

DROS-Kinetic provides deterministic containment only within the stated system and disturbance assumptions. It does not protect against physical flight controller hardware destruction, sub-transceiver physical wiretapping, or atmospheric wind gusts exceeding the aerodynamic thrust envelope ($W > W_{\text{max}}$).

TABLE I
Boundary-Oriented Robustness and Stress Metrics for Autonomous UAV Physical Execution

Workload Metric	Evaluated Workload			Baseline (B0 - Unprotected)	DROS-Kinetic (B1 - Guarded)	Status
Unauthorized Containment (R_{contain})	In-Flight Injection	Disarm	Injection	0% (Motor Cutoff / Crash)	100% (Stable Hover Maintained)	Pass
Fail-Closed Rate (R_{FC})	10,000/10,000	Mutations		N/A (Unhandled Exceptions)	1.000 (Deterministic Lockout)	Pass
Engine Crash Count	10,000 Tests	Boundary		> 120 Daemon Faults	0 Crashes (Zero Memory Leak)	Pass
Lookahead Stop Margin	15.0 m/s	High-Speed Cruise		-180.0 m (Boundary Breach)	+24.60 m (Safe Stopping Distance)	Pass
Enforcement Latency P_{50}	In-Band	Micro-benchmark		N/A	4.40 μs (L_2 Execution Gate)	Pass
Enforcement Latency P_{99}	178,000	QPS	Stress Load	N/A	13.30 μs (L_2 Jitter Bound)	Pass

B. Generalization to Other Physical AI Platforms

The mathematical invariants formalized in this work extend naturally to other embodied AI systems:

- Humanoid Robotics: Joint torque clamping and velocity envelope restriction ($\Delta\tau_{\text{cmd}} \equiv 0$).
- Autonomous Vehicles: Drive-by-wire steering and emergency braking horizons.

IX. Conclusion

This paper addressed the Post-Compromise Physical Action Authority Problem in Physical AI. By decoupling cognitive controller integrity from physical execution authority, we formalized the UCIAI invariant and conditional safety-envelope preservation. Through SITL quadrotor experiments across the T1–T6 protocol, we demonstrated that in-band runtime enforcement can deterministically prevent unauthorized physical actuation even when high-level AI reasoning is entirely subverted.

Acknowledgment and AI Declaration

Generative AI tools were used solely for technical writing assistance, grammatical refinement, and $\text{\LaTeX}$ typesetting. The core conceptual vision, research questions, system architecture, formal invariants, physical causality models, experimental methodology, analysis of results, and claims were independently formulated and verified by the authors. The authors retain full accountability for all content presented in this work.

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